

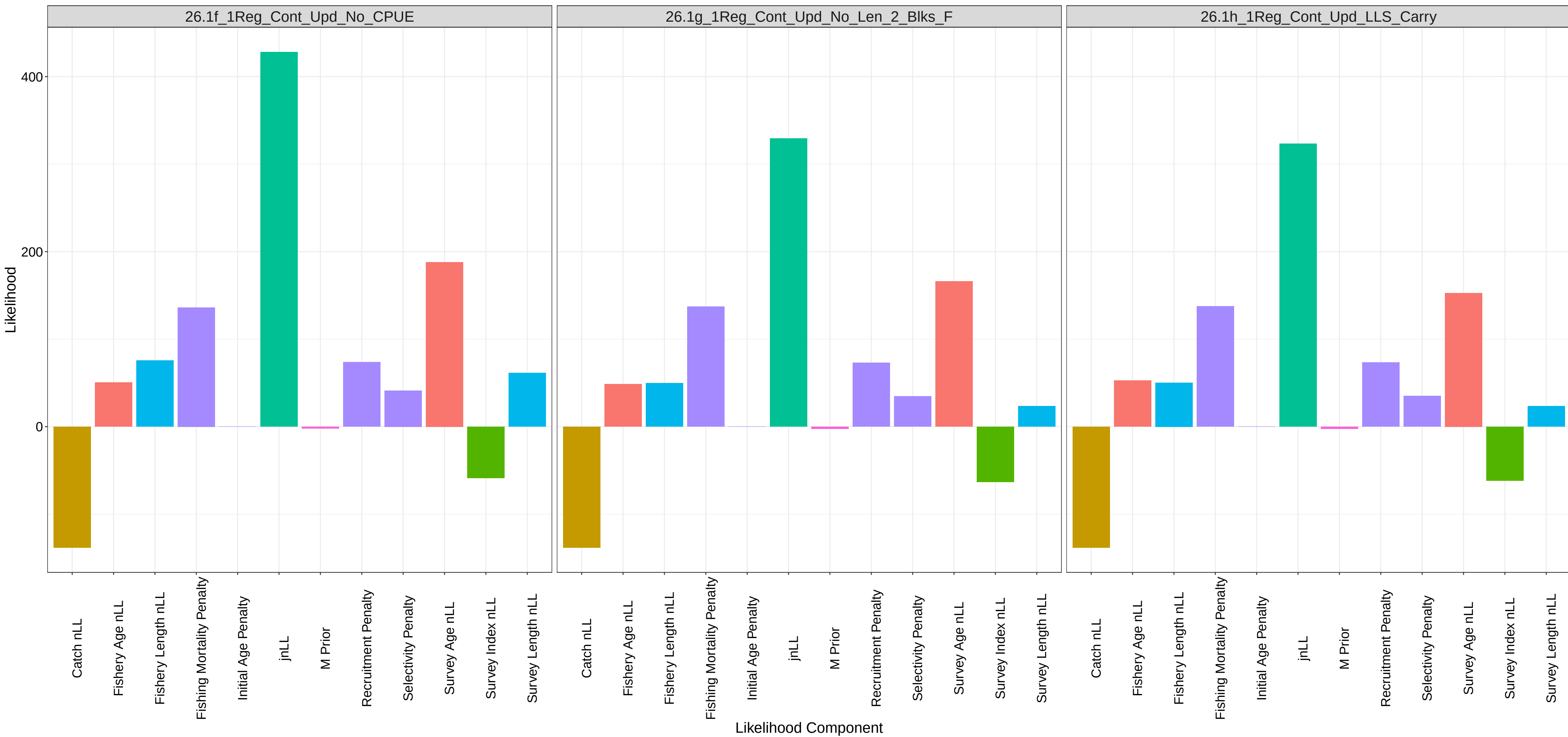
	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	26.1h_1Reg_Cont_Upd_LLS_Carry	1	230.4754	382.9105	0.04375466	43.52004	142.4181	0.07947	1.61830	0.60190	0.55056
2	26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F	1	204.7788	360.0882	0.04872276	38.40136	135.5328	0.07778	1.51092	0.56869	0.62644
3	26.1f_1Reg_Cont_Upd_No_CPUE	1	230.2295	384.2490	0.04435560	42.49986	143.0228	0.07854	1.60974	0.59917	0.56478

	Model	jnLL	M Prior	Recruitment Penalty	Selectivity Penalty	Catch nLL	Fishing Mortality Penalty	Survey Index nLL	Initial Age Penalty	Survey Age nLL	Fishery Age nLL	Survey Length nLL	Fishery Length nLL
1	26.1h_1Reg_Cont_Upd_LLS_Carry	323.4643	-2.290555	73.40731	35.23643	-138.2941	137.6949	-61.68283	0.1663095	152.7626	52.76715	23.36413	50.33307
2	26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F	329.2878	-2.498959	73.03328	34.80583	-138.2997	137.2802	-63.32470	0.1809585	166.2168	48.68255	23.53528	49.67626
3	26.1f_1Reg_Cont_Upd_No_CPUE	428.1021	-2.132048	73.91040	41.32049	-138.1116	136.2327	-58.77612	0.1697382	188.0172	50.39396	61.34404	75.73329

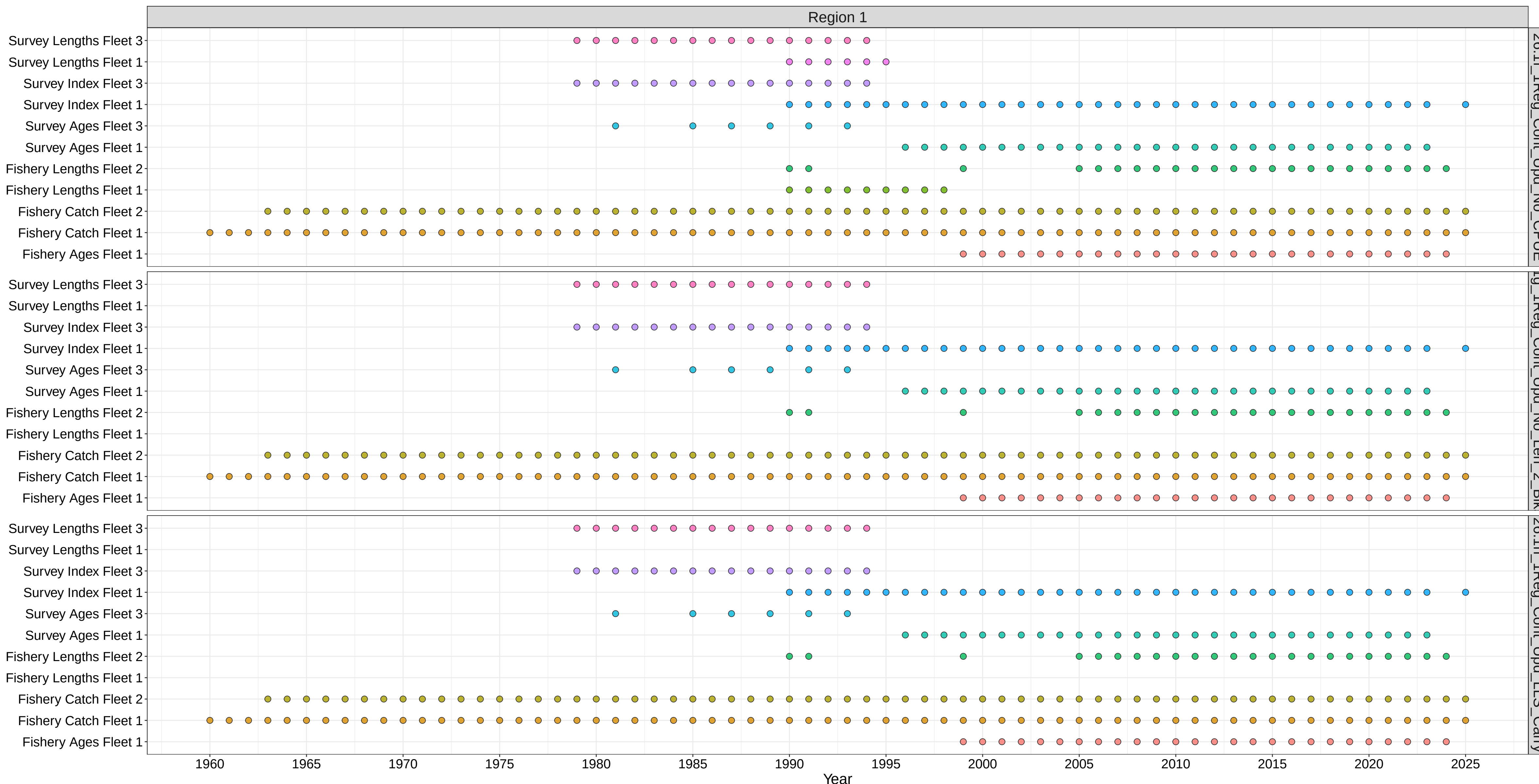
Negative Log-Likelihood

Type

Age	Index	Length	Prior
Catch	jnLL	Penalty	

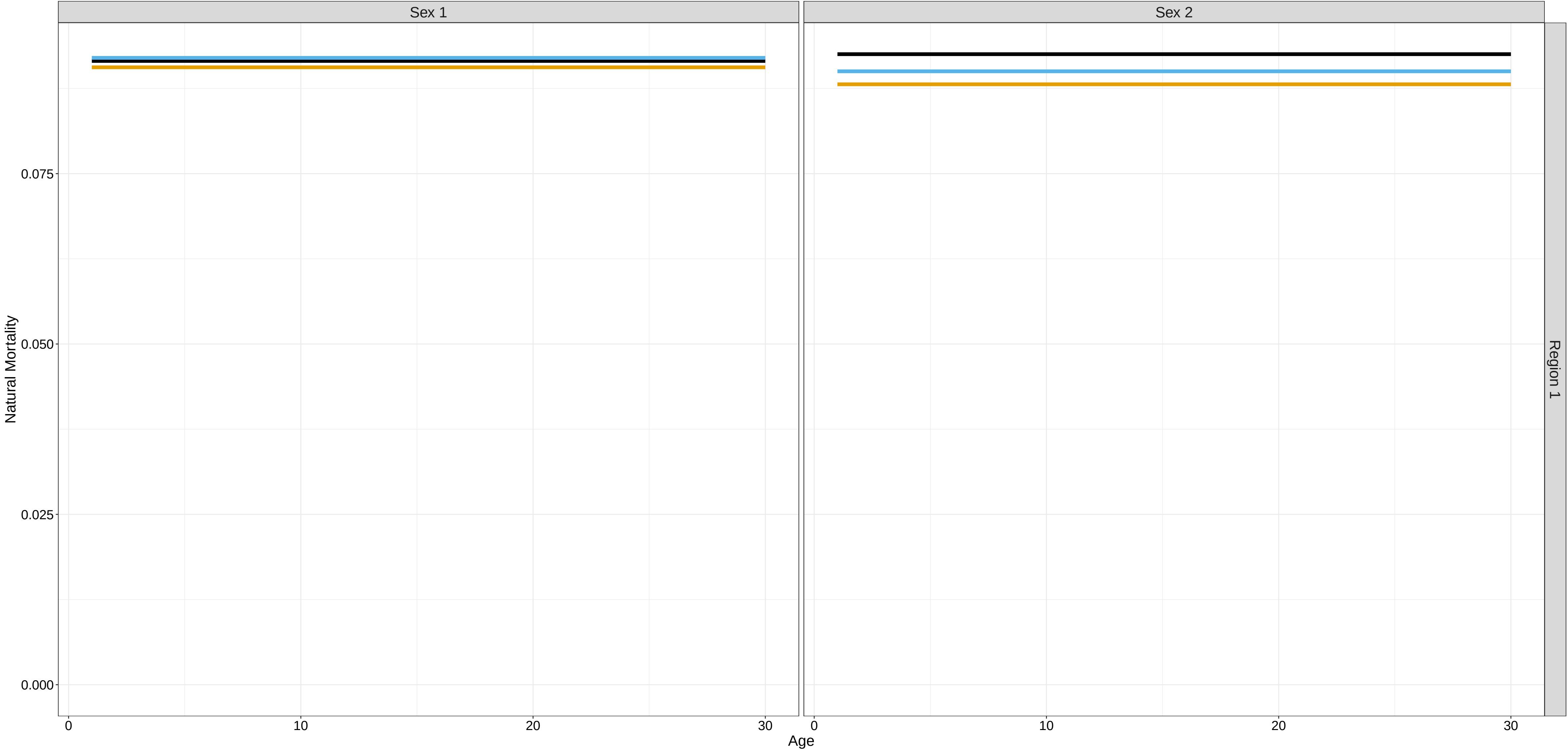


Data



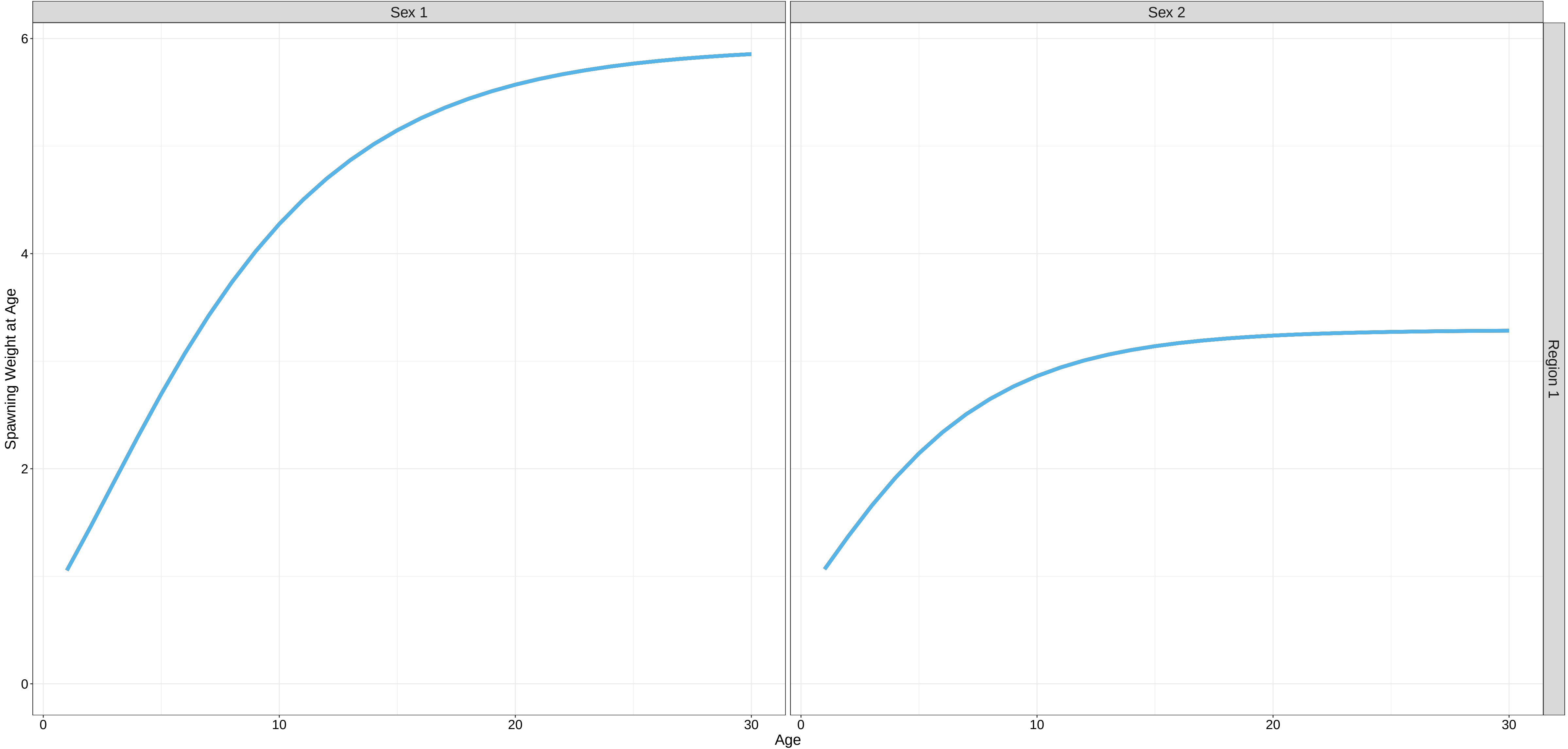
Natural Mortality

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



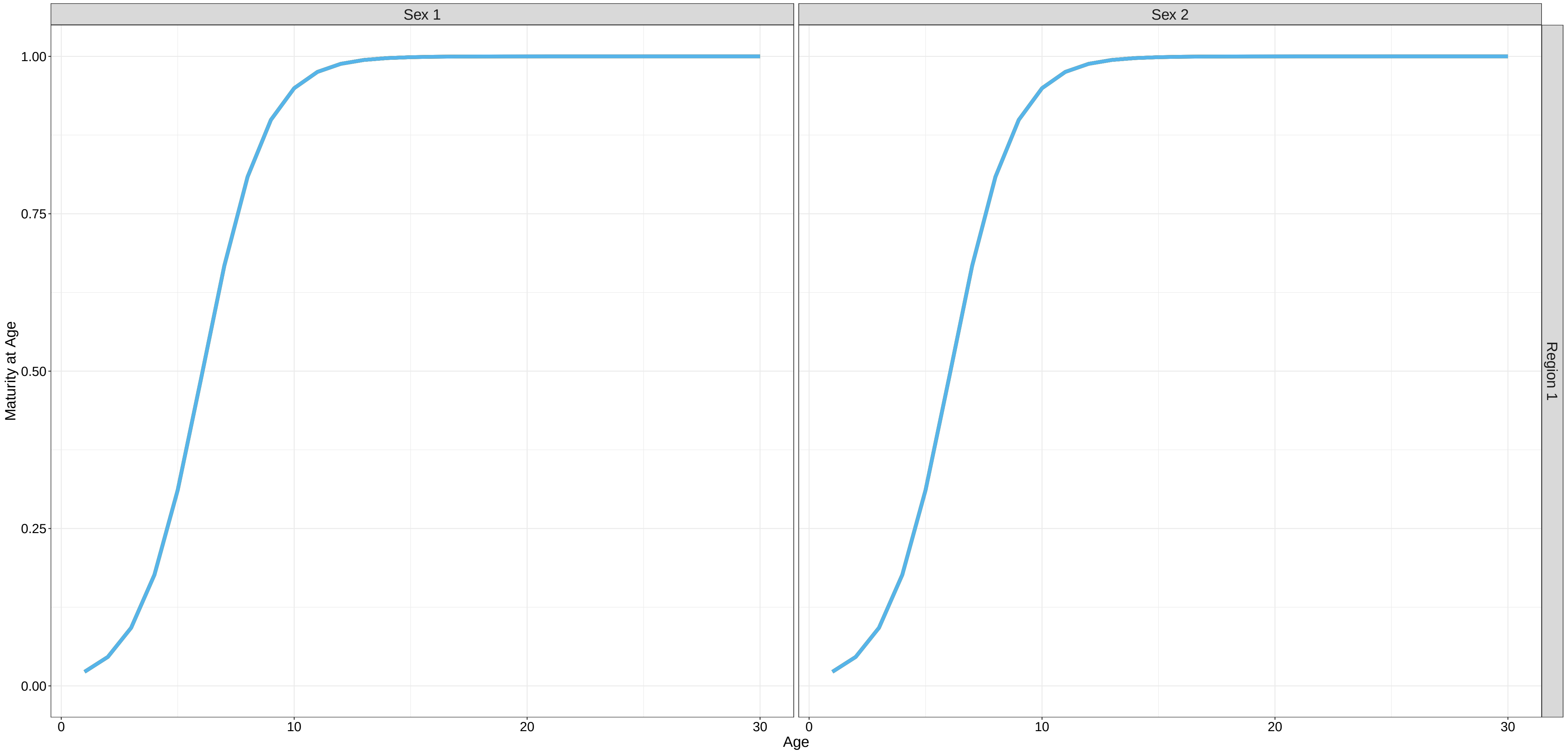
Weight-at-Age

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



Maturity

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry

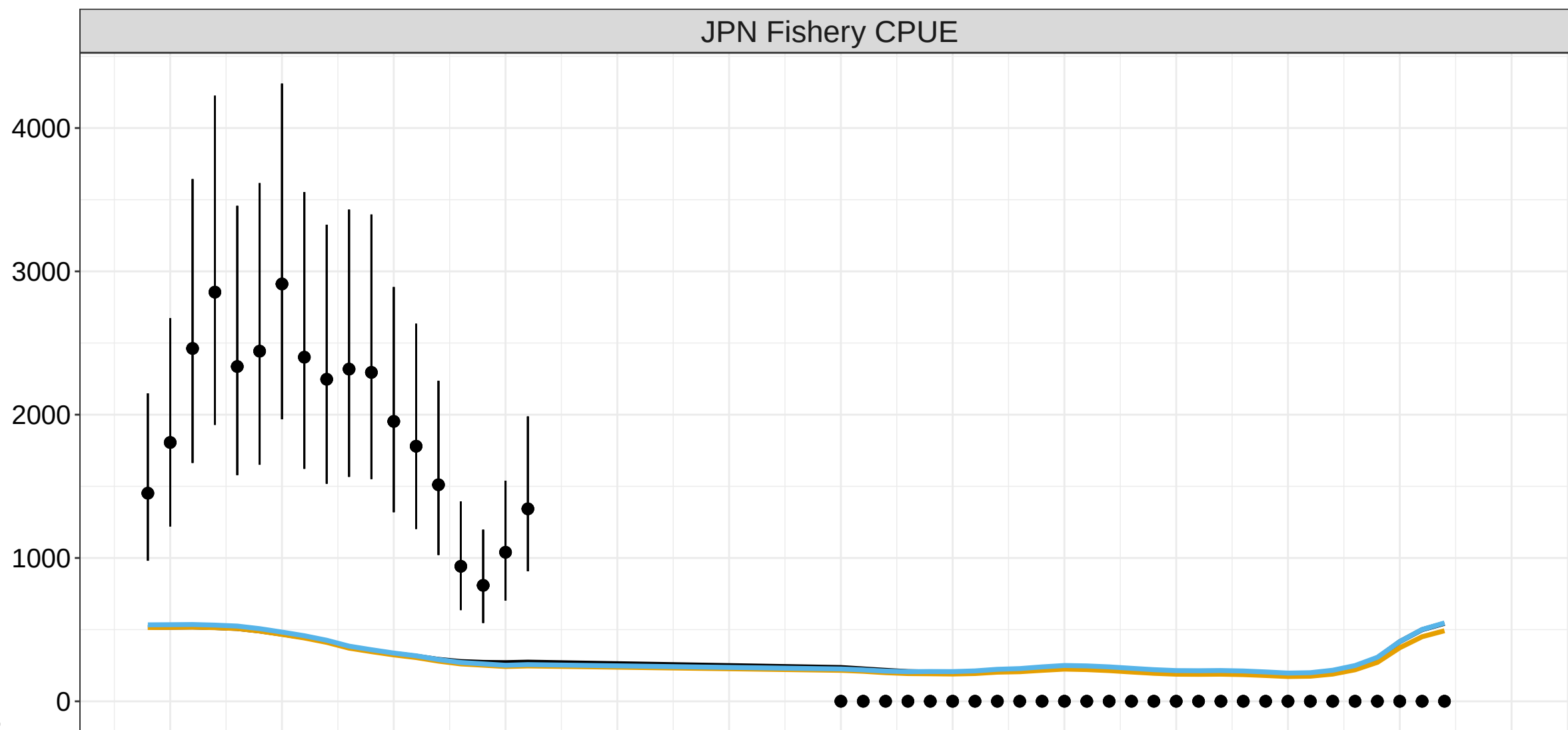


Index Fits

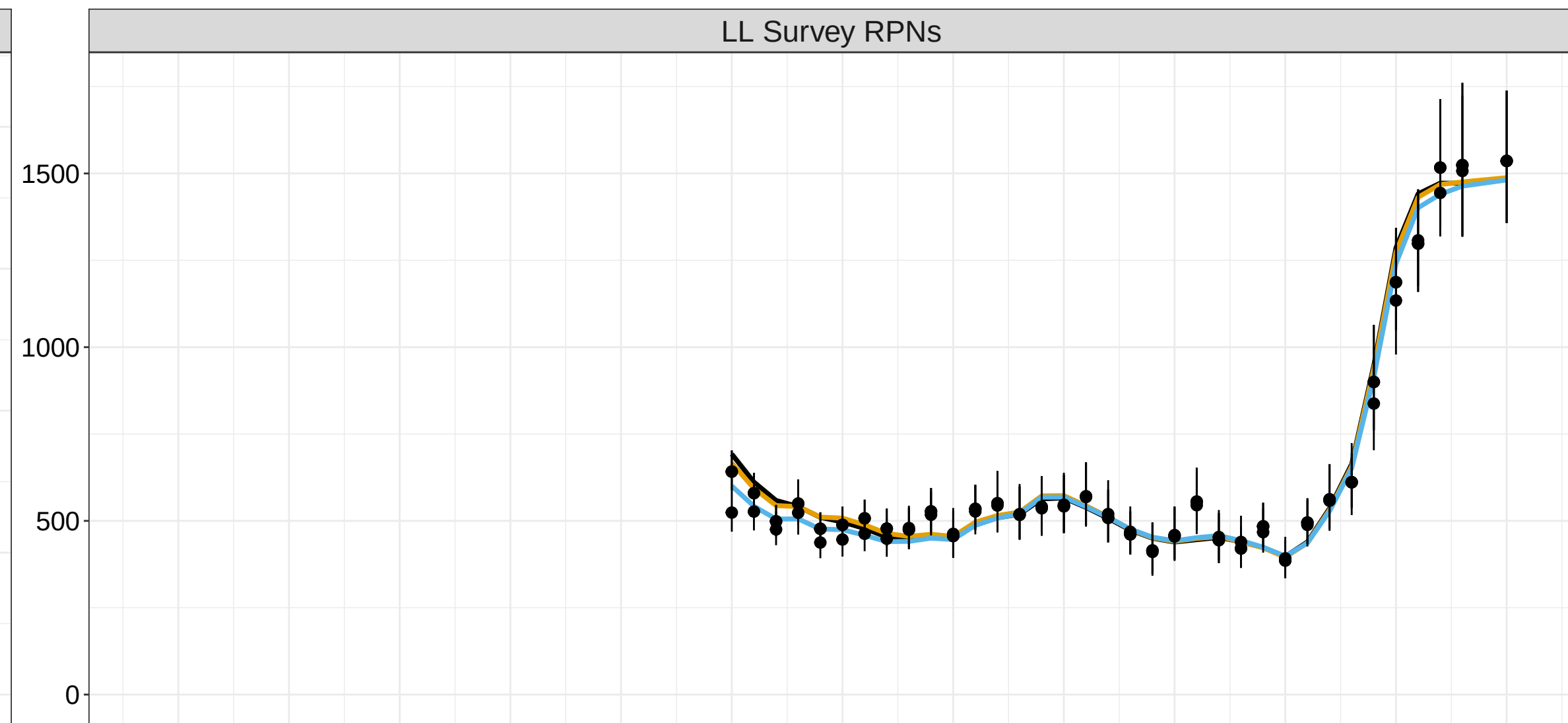
Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry

Index

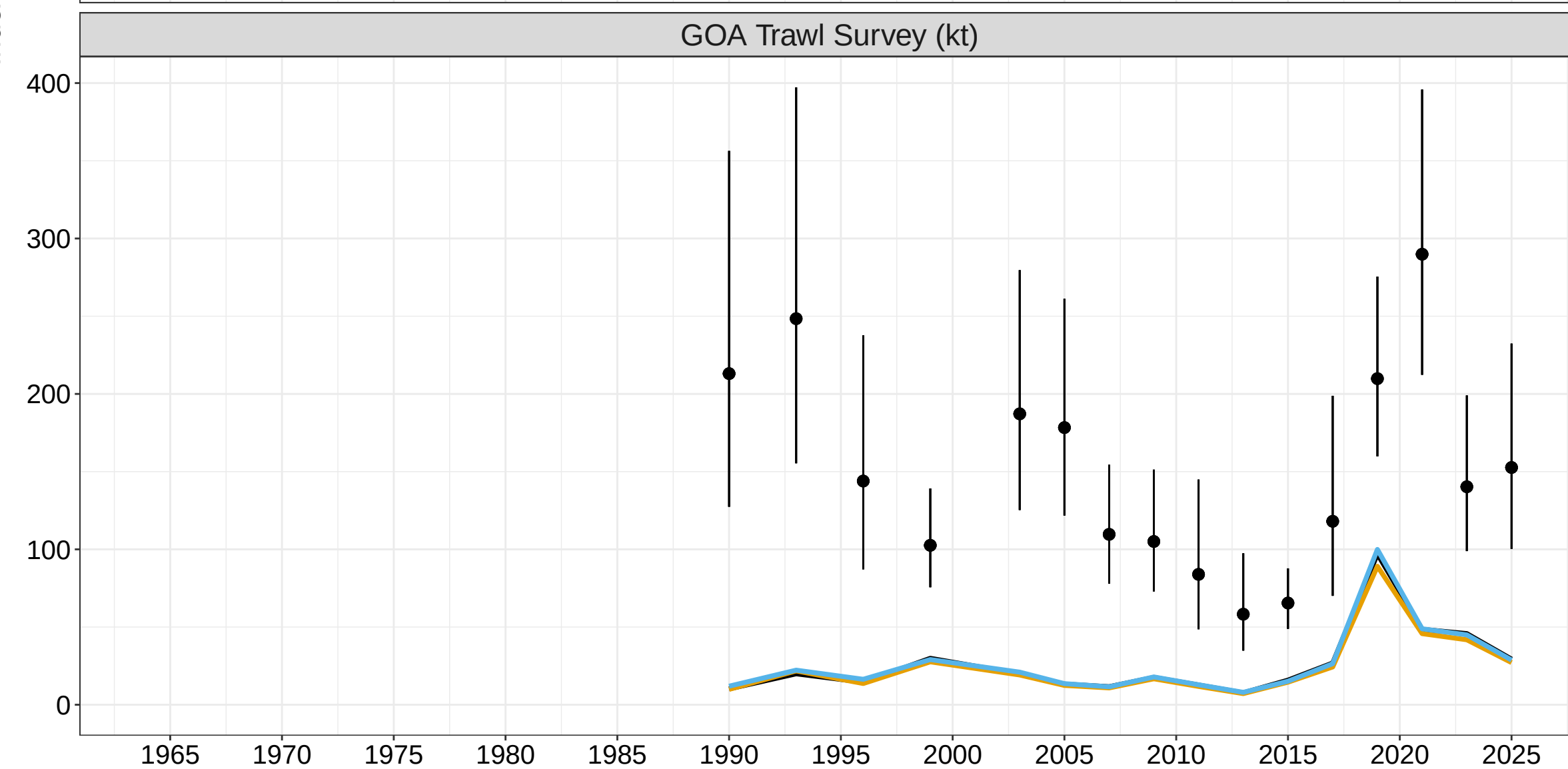
JPN Fishery CPUE



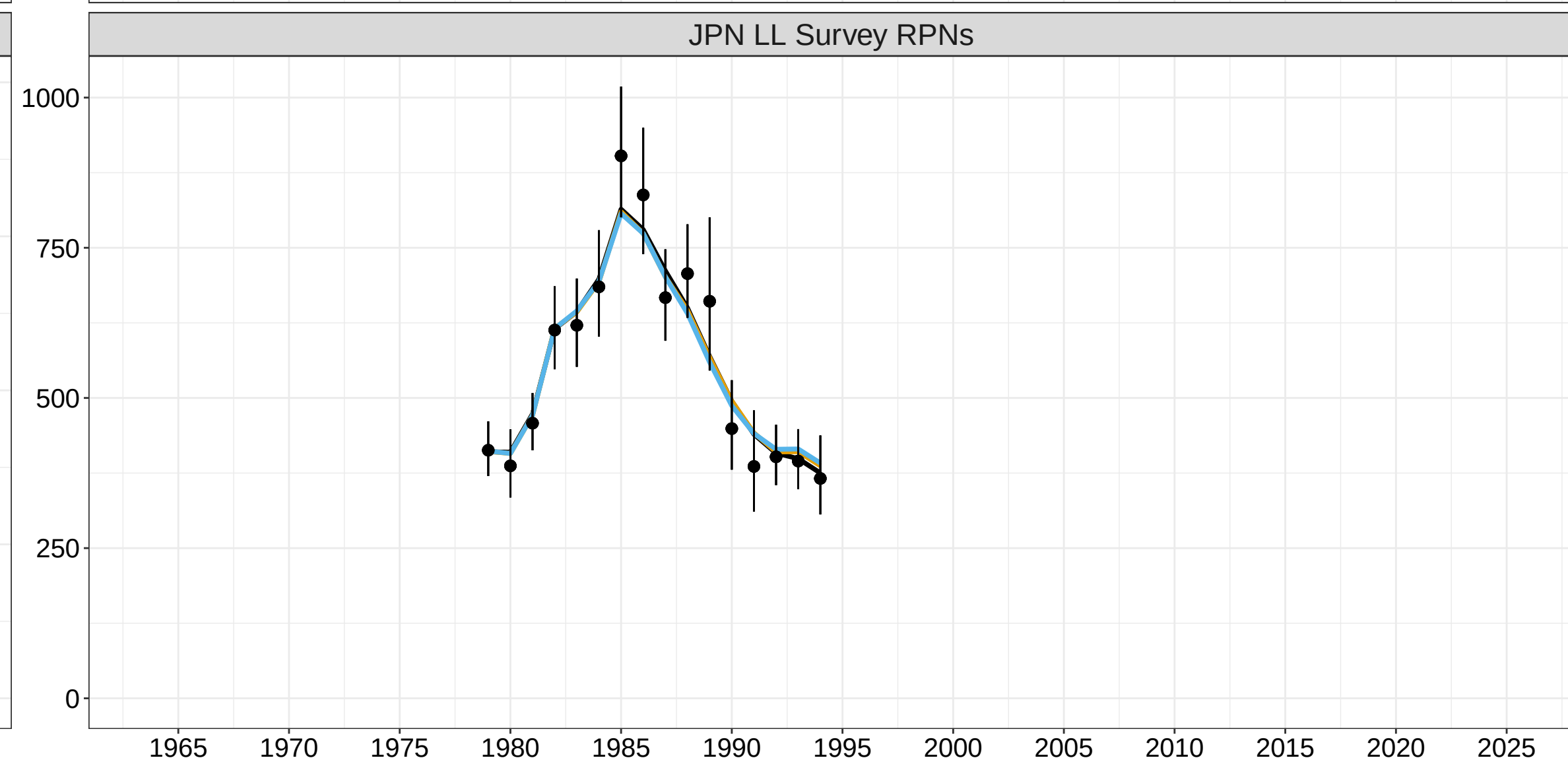
LL Survey RPNs



GOA Trawl Survey (kt)



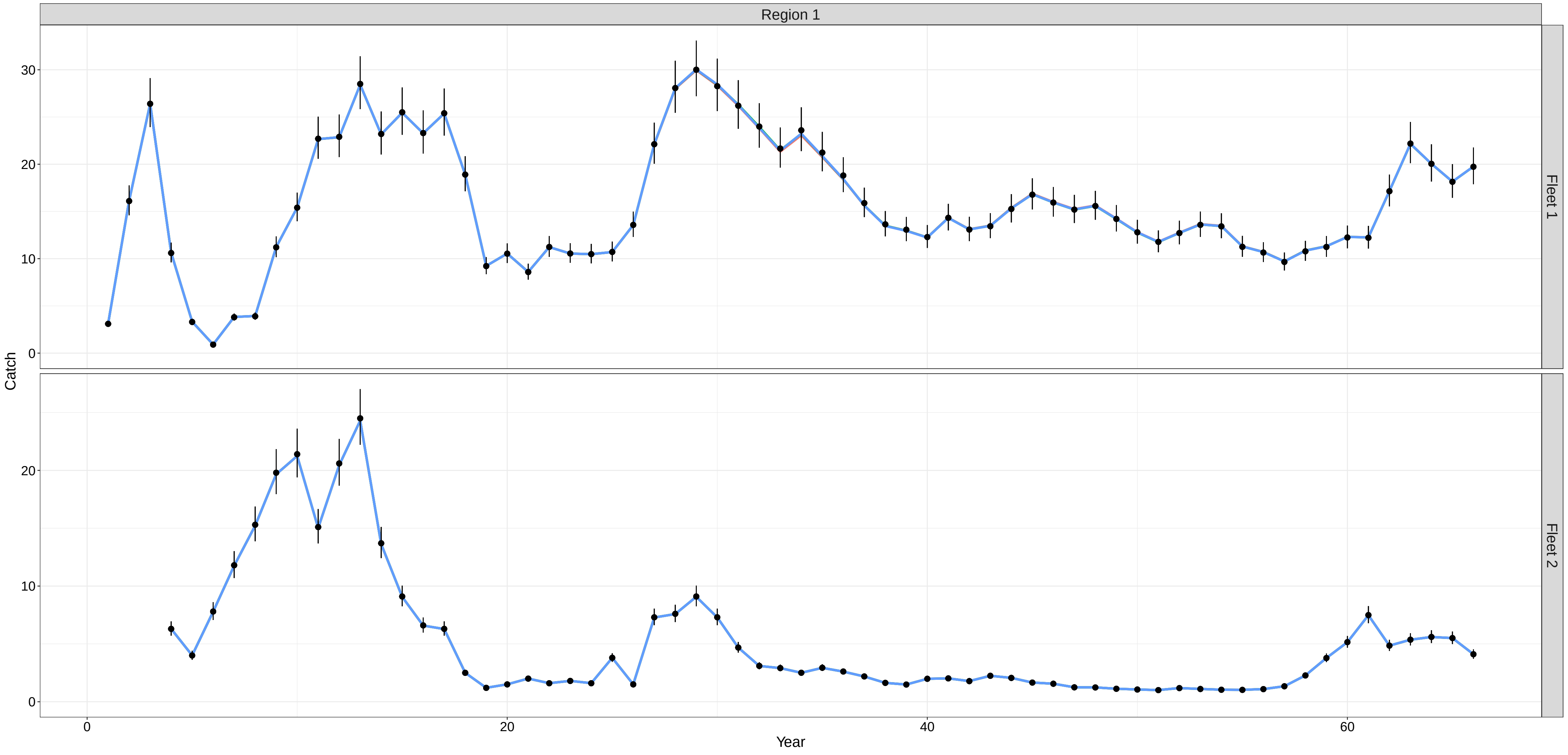
JPN LL Survey RPNs



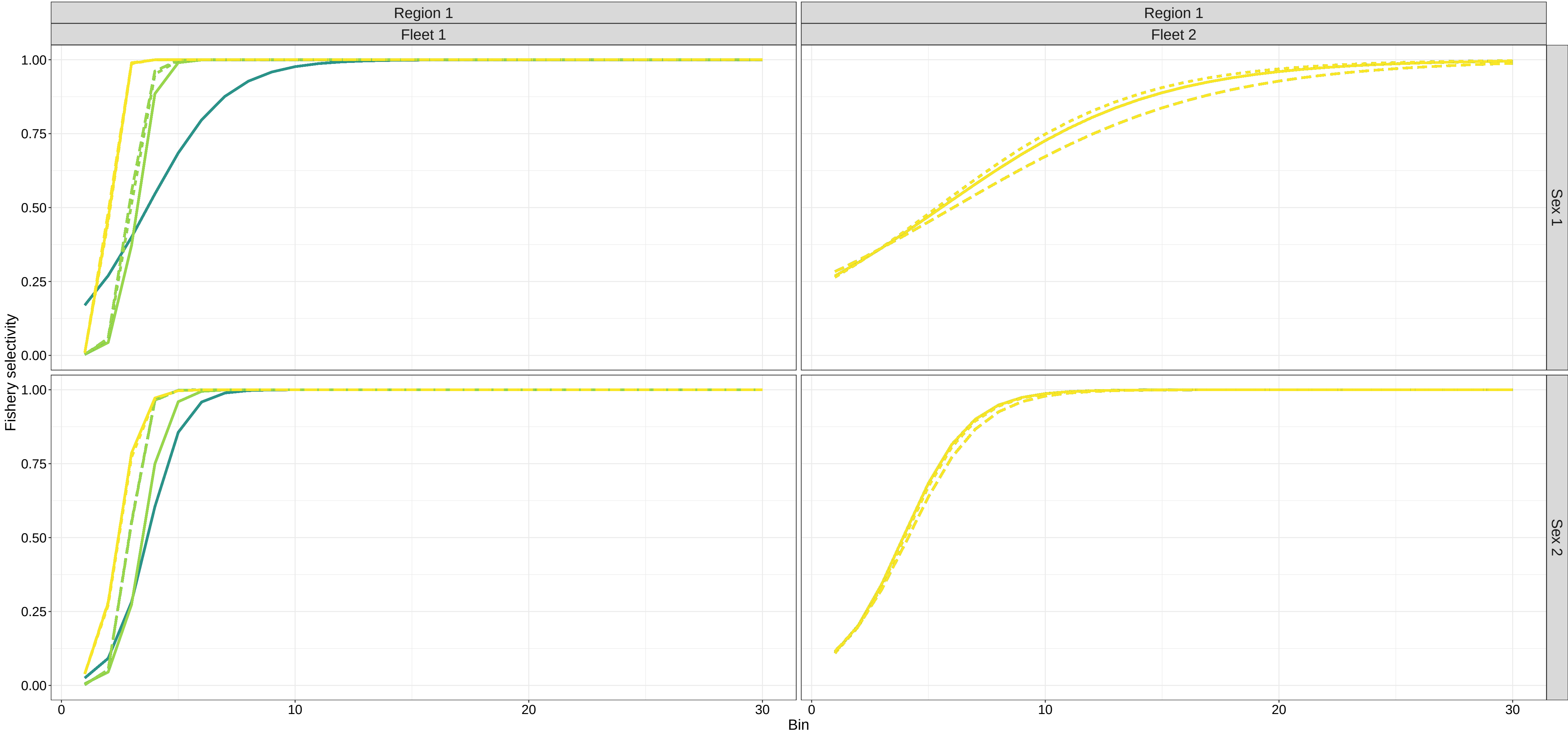
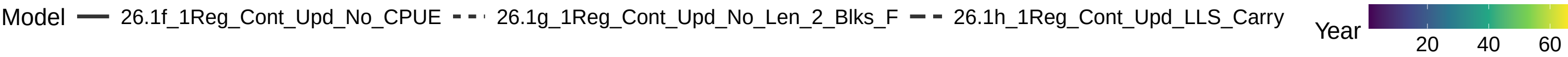
Year

Catch

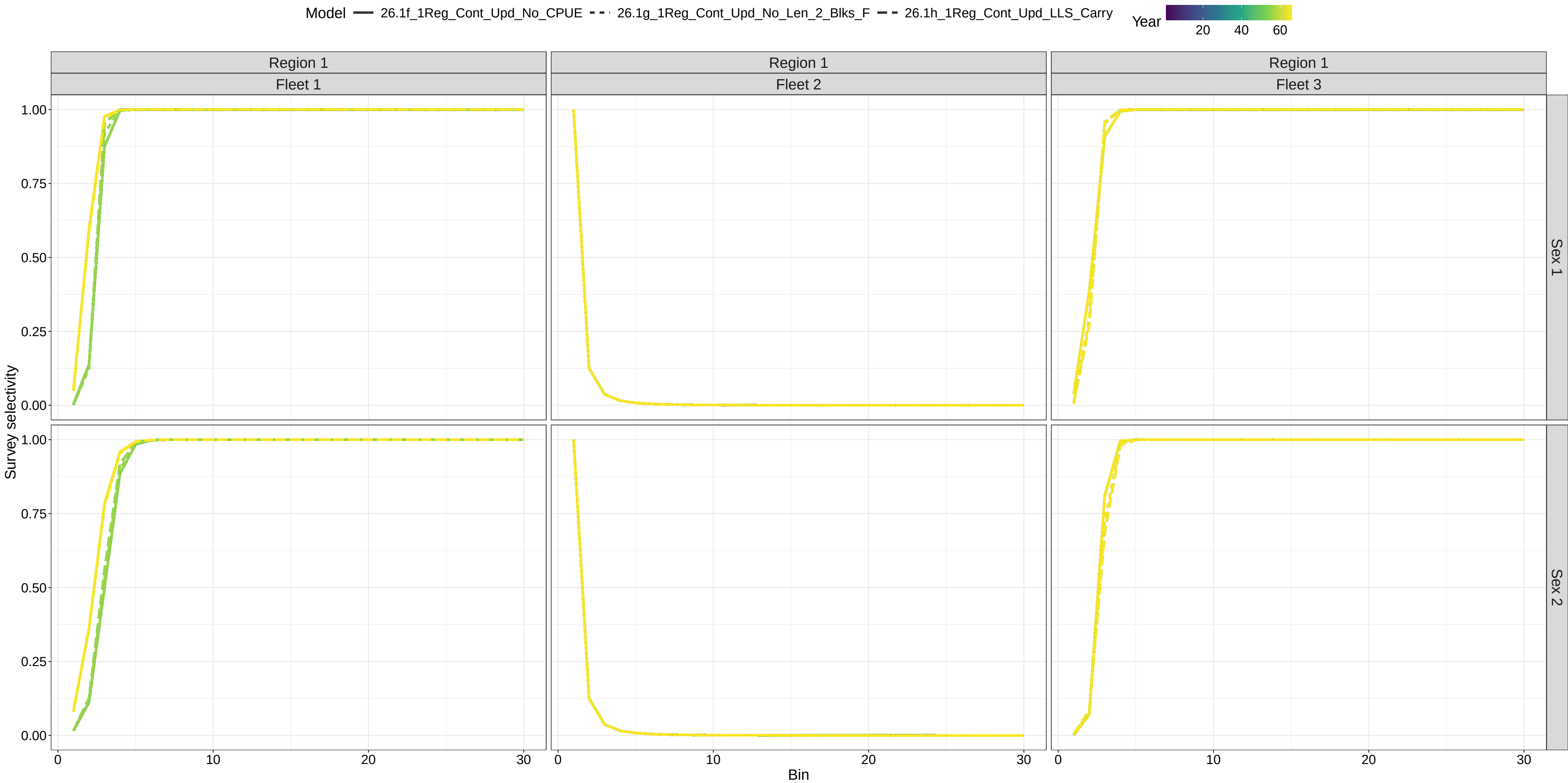
Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



Fishery Selectivity

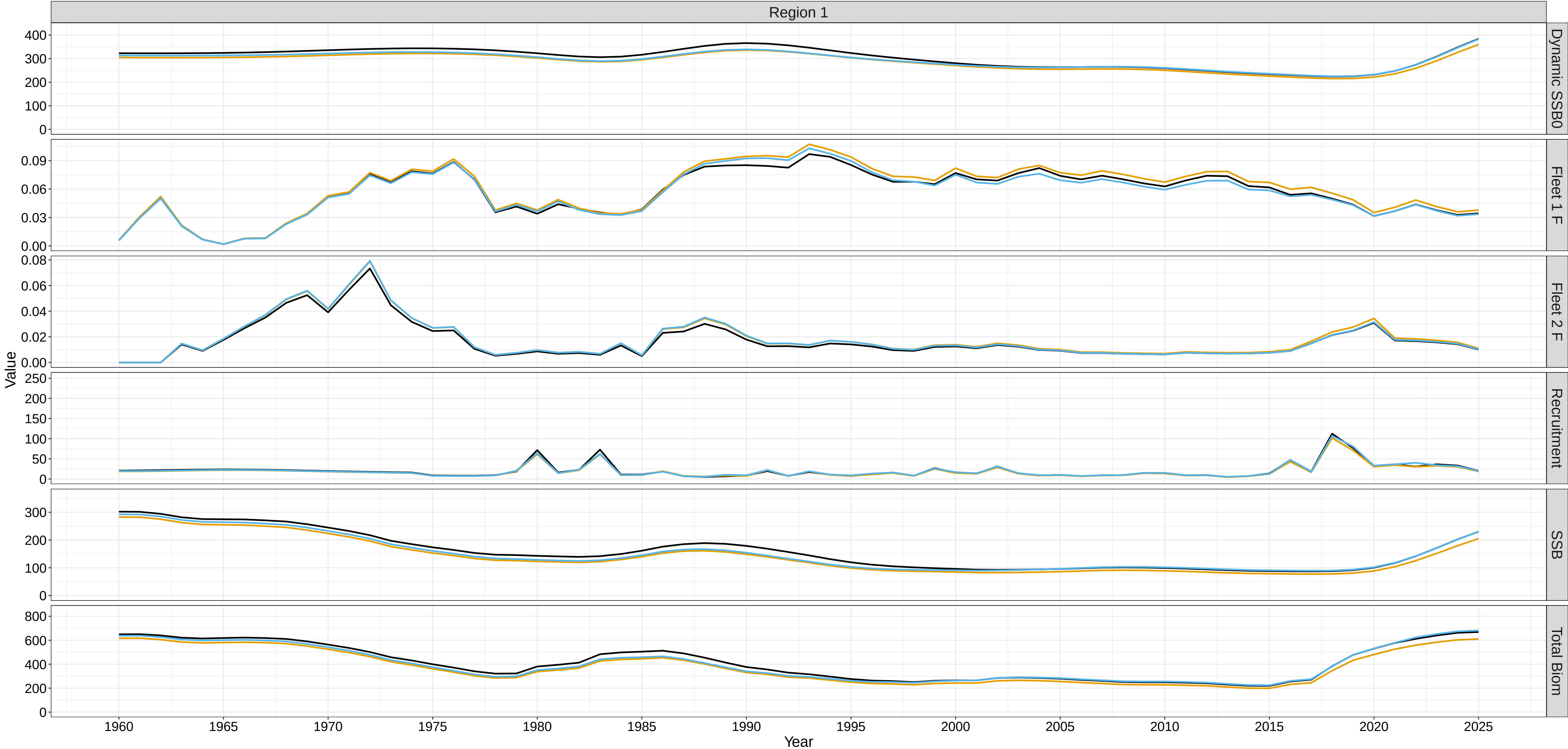


Survey Selectivity



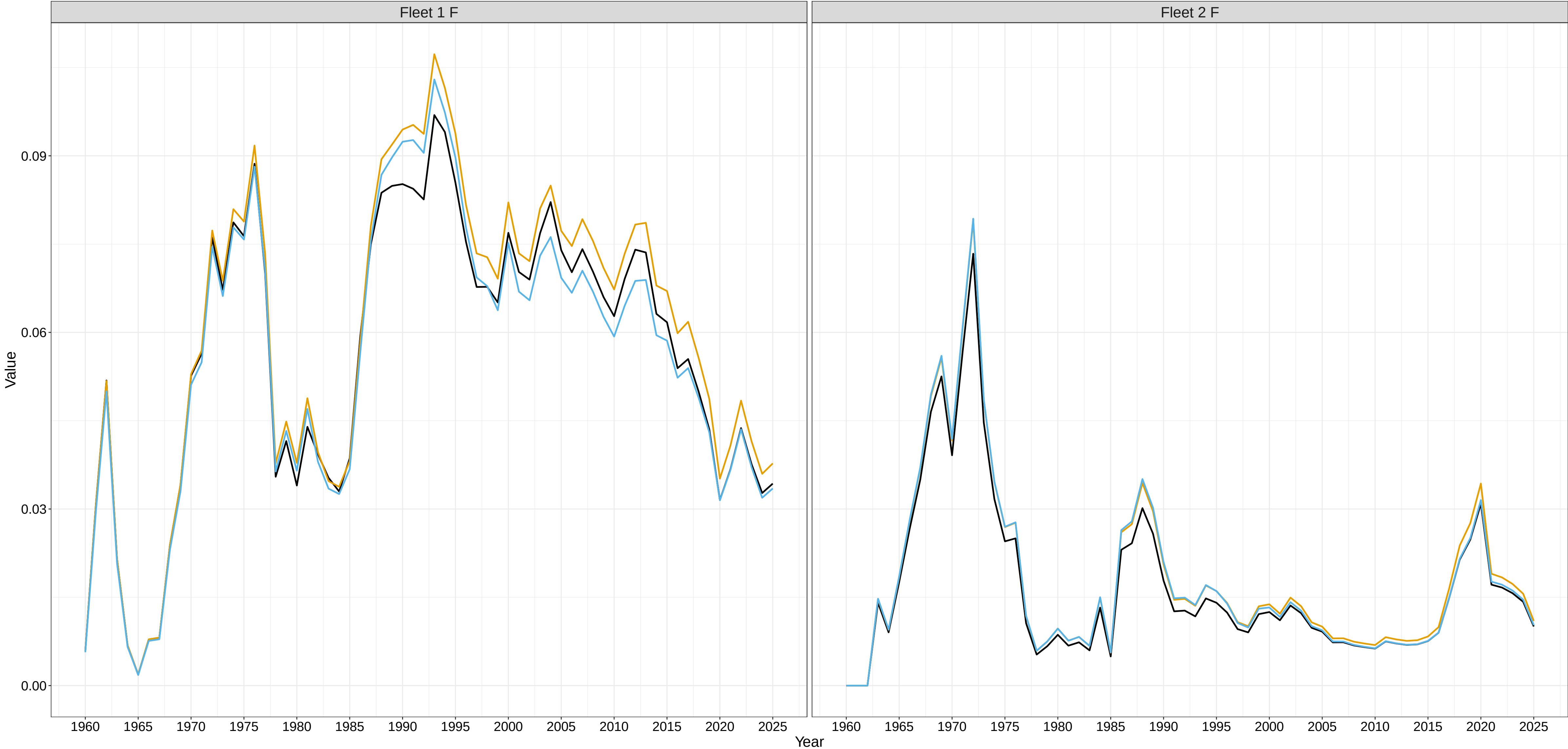
Time Series Plots

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



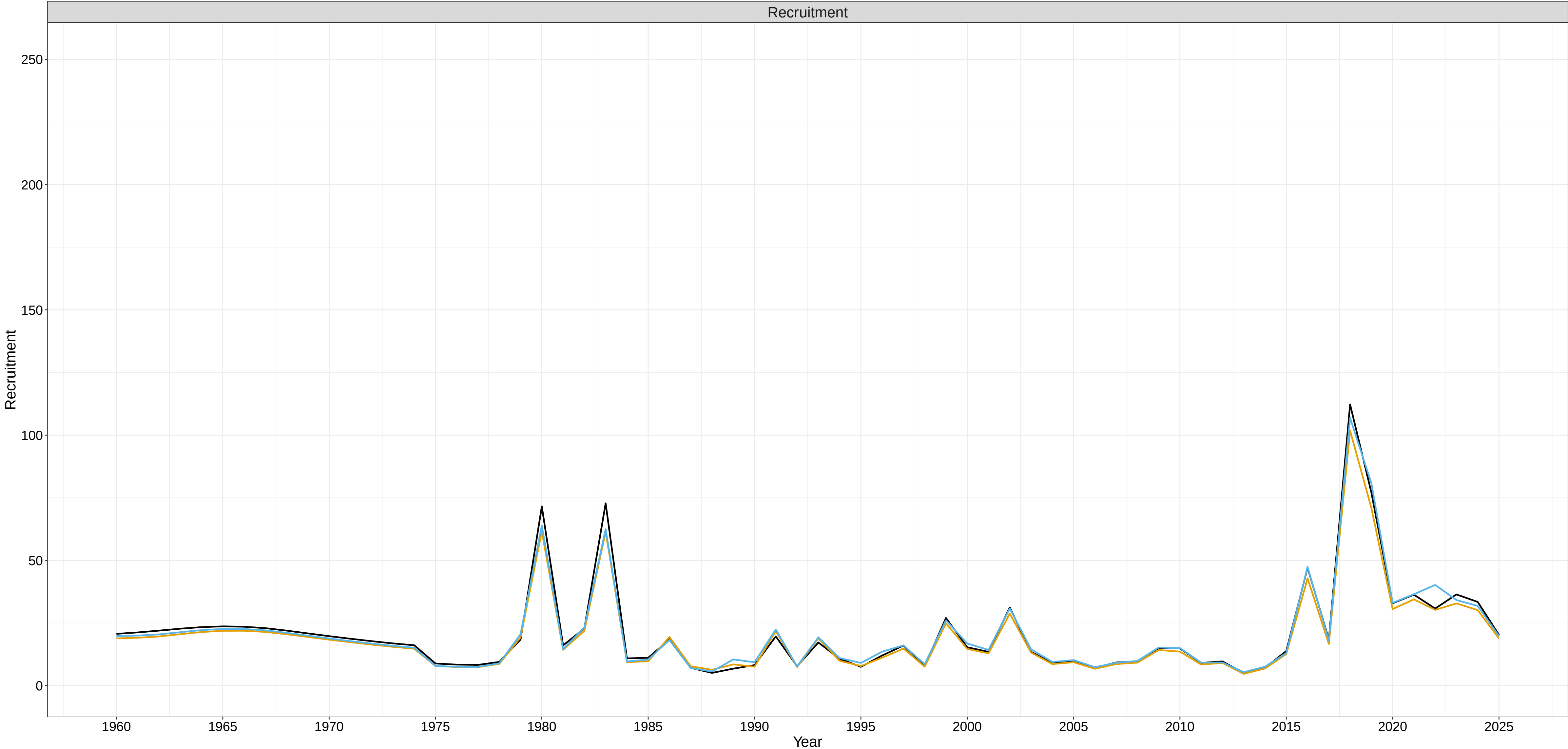
Fishing Mortality

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



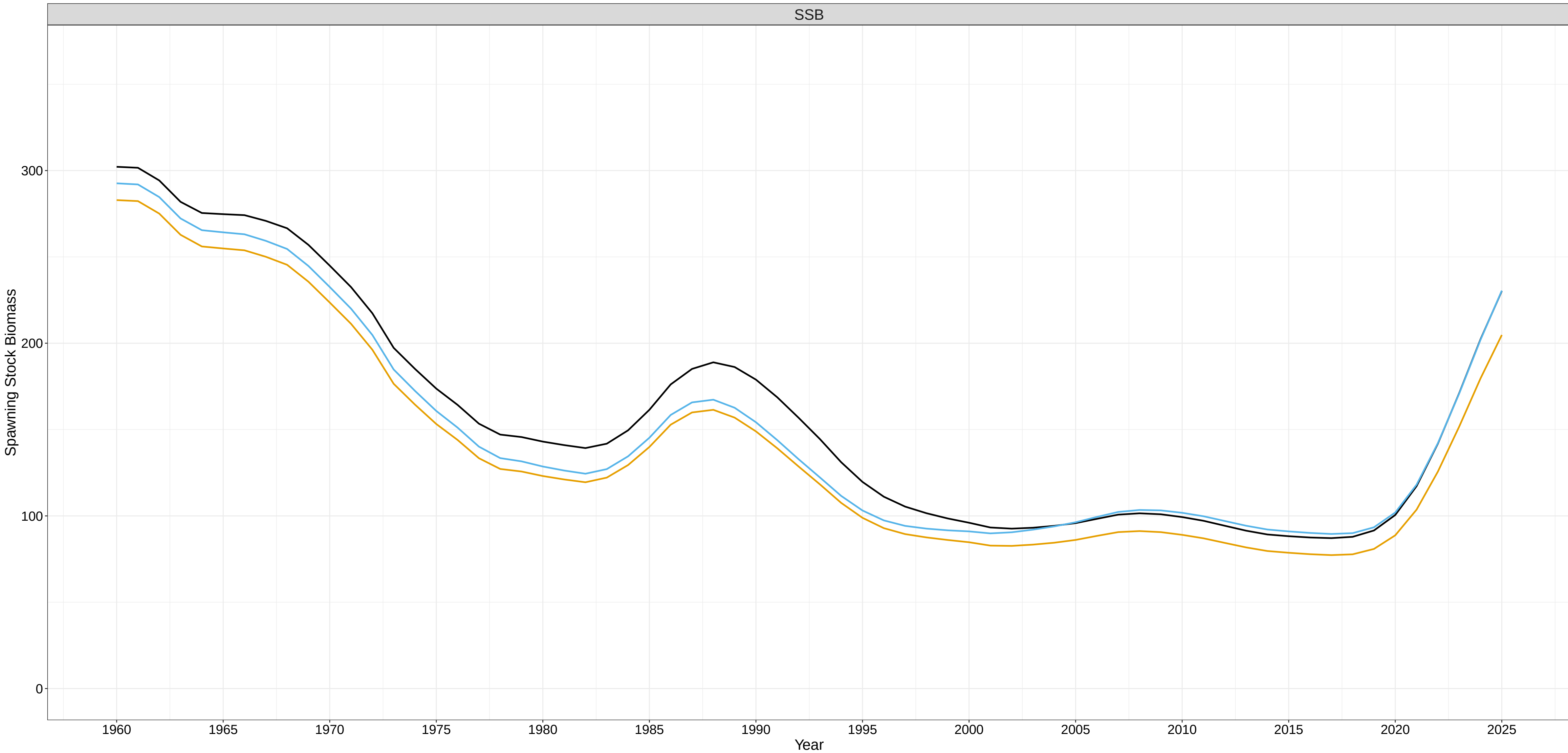
Recruitment

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



SSB

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



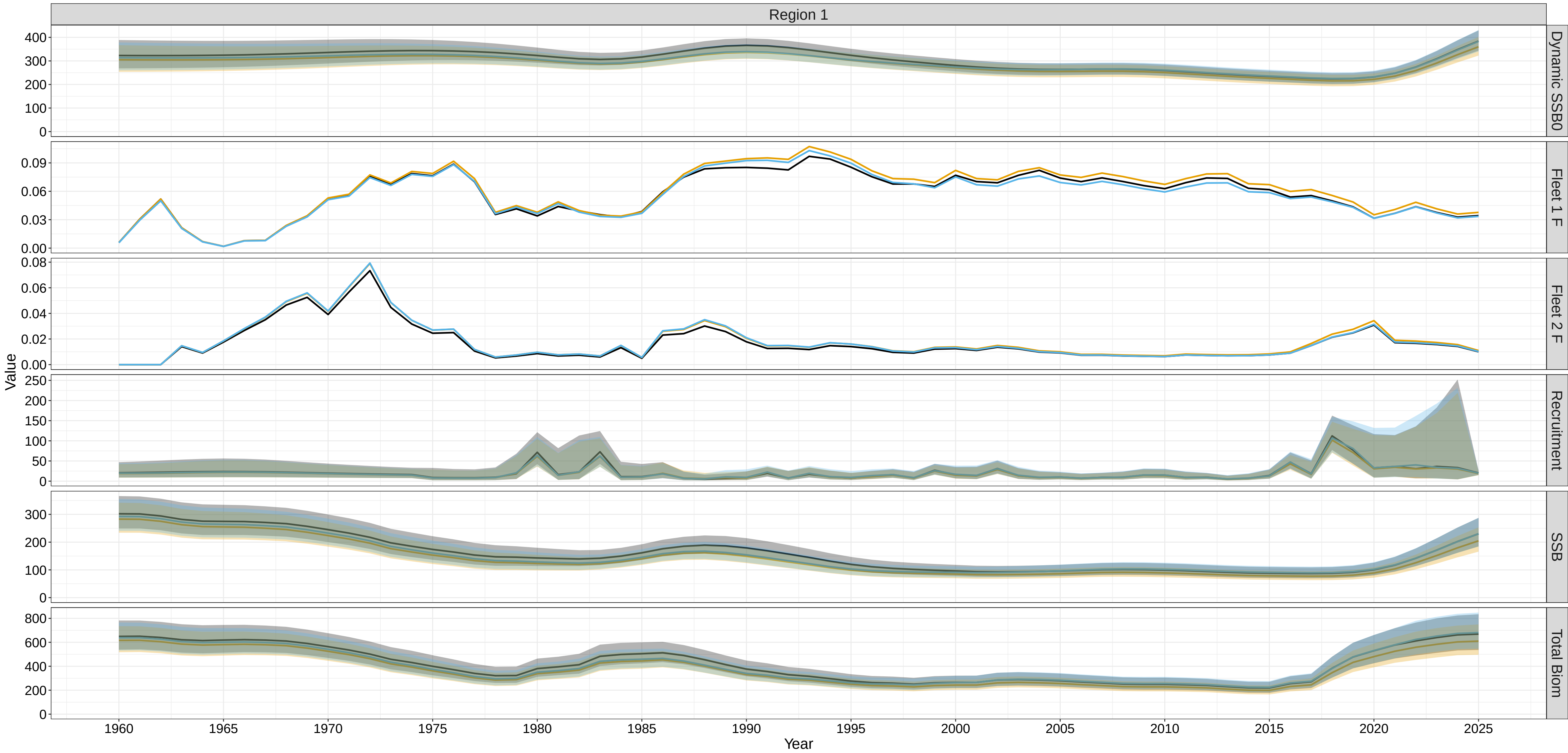
Biomass

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



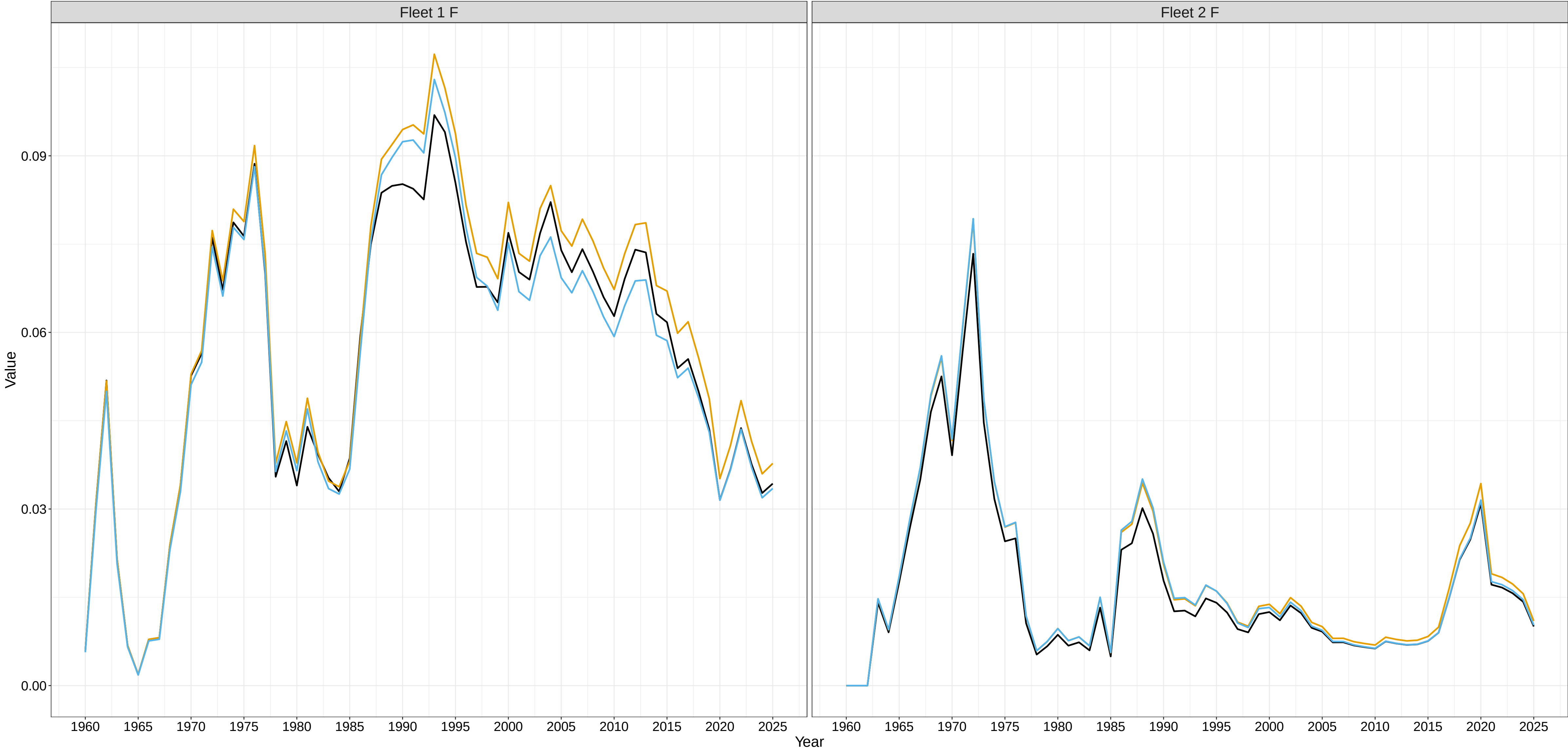
Time Series Plots

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



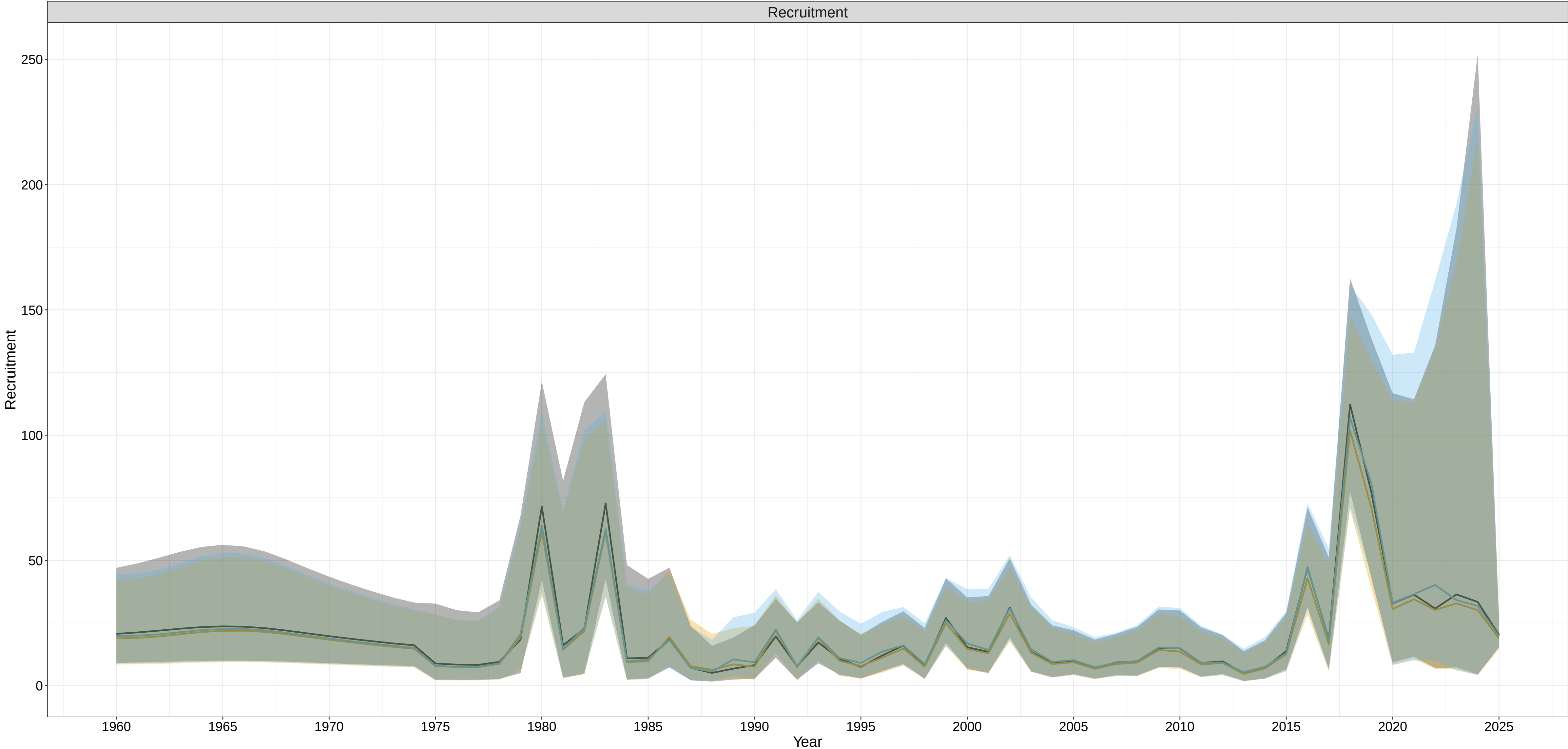
Fishing Mortality

Model — 26.1f_1Reg_Cont_Upd_No_CPUE — 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F — 26.1h_1Reg_Cont_Upd_LLS_Carry



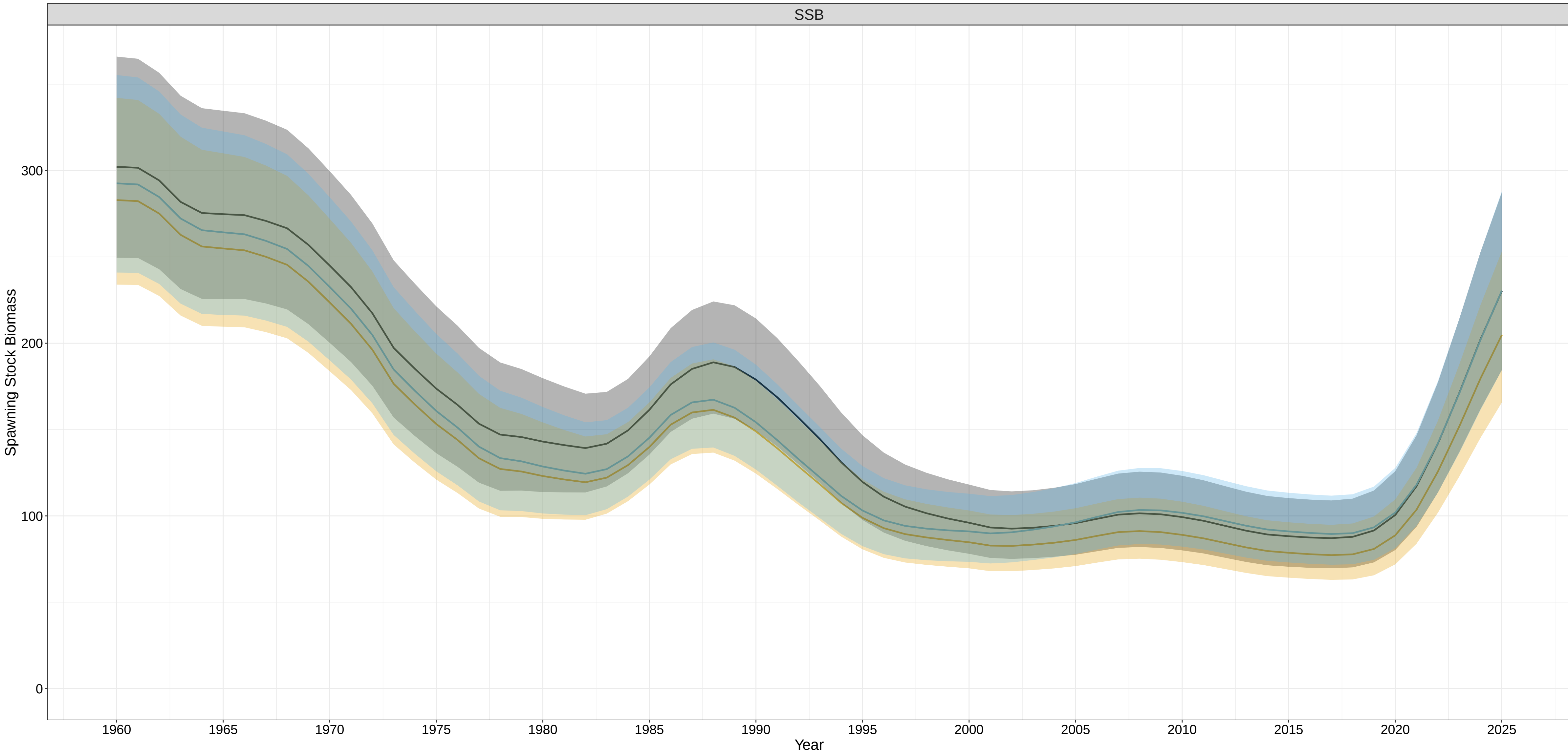
Recruitment

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



SSB

Model 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry



Biomass

Model — 26.1f_1Reg_Cont_Upd_No_CPUE 26.1g_1Reg_Cont_Upd_No_Len_2_Blks_F 26.1h_1Reg_Cont_Upd_LLS_Carry

